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RTES →
{COMMUNICATION}
→ TCP/IP protocol Suite

→ TCP Transmission control protocol
IP – Internet protocol.

* It is basic communication protocol used for the internet & Most computer N/w

→ IP is responsible for
- Assigning IP address
- Sending data from 1 device to other
- Routing data across N/w

- TCP
- Breaking data in to Small packets
- Ensuring data is delivered correctly
- Recording packets
- Resending lost packets

- How it works →
→ Data is devided into packets (TCP)
→ Each address packet get an adress (IP)
→ Packet travel through N/w
→ Reciever reassemble Packet (TCP)

→ There are 5 layers in TCP/IP

→ APPLICATION LAYER

→ Transport layer

→ Internet layer

→ Data Link layer

→ PHYSICAL LAYER

→ PHYSICAL LAYER -

This layer define characterstics of the transmission as data rate & Signal encoding scheme.

Exa - transmitting data or connecting two device through
- Twisted pair cable
- fiber optic cable.

→ DATA LINK LAYER

- combines Data link + physical function
→ It is Responsible for
* Framing * Mac addressing
* Error detection * physical transmission of data

→ Example → Ethernet + WIFI

→ This ~~layer~~ defines ~~the~~ protocol ~~to~~ ~~manage~~

→ In LAN, The data link layer ID devided in Two Sub type

- Medium access control (MAC) Sub layer →

- lower Sub layer of Data Link layer
- control how devices access the physical medium.

- Logic link control (LLC)

- Upper Sub layer of Data link layer
- Act as interface ~~b/~~ N/w layer & MAC

3→ Internet LAYER →

- Responsible for logical addressing & routing
- Decide the best path for Data transmission
→ Most of embedded System use 32 bit Protocol →
- IP - Assigning IP addresses
- ICMP → Error reporting
- ARP - converts IP to MAC address

→ present running IP layer Software is called IPV4 this version will slowly Replaced by IPV6
- IP V_4 is 32 bits → IPV6 → 128 bits.

3] → Transport layer →

- Provides end to end communication
- Responsible for
  - Segmentation of data
  - Error detection
  - Flow control
  - Port addressing
- Transport layer software run on every end
- Main protocol
  - TCP,
    - Reliable – connection-oriented
    - Ensure data delivery
  - UDP,
    - Faster – connectionless
    - No gurantee of delivery

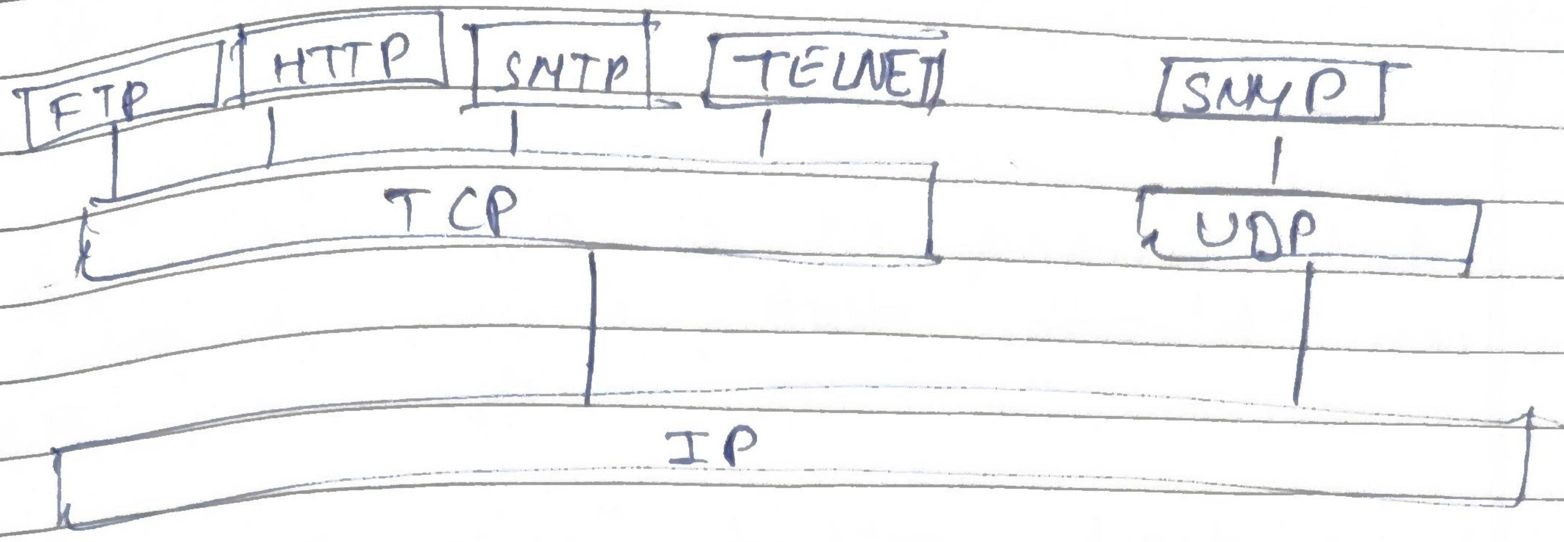
5] APPLICATION LAYER –

- Topmost LAYER of TCP/IP Model
- Provide services directly to user application
- Handle data formatting, encryption & session management

→ This application differ from application to application

→ The data link layer frame is sent to the physical layer interface & then bit stream is sent over the transmission medium

APPLICATION LAYER protocol in TCP/IP protocol stack ->



- SMTP -> Simple mail transfer protocol
  use for Sending mail

- FTP - File transfer protocol
  for file transfer

- HTTP -> HYPER Text Transfer protocol
  for world wide web commiwkation

- ~~DNS~~ -
- MIME -> multimedia Internet mail extension
  - for electronic mail with multimedia content
- TELNET use for Remote login

- SNMP - Simple N/w management protocol
  for N/w management

TASK SCHEDULING

Introduction →

- Task scheduling is the process of deciding when & which task should use CPU in System

- In real time system, task must complete before their deadline.

Task can be :
- Periodic – Repeat at fixed intervals
- Apeordic/ sporadic – occur randomly or happen at unpredictable time

- The main goal is to ensure all task meet deadline.

Process utilization –
- Process utilization shows how much CPU time is used

- For one task –
  utilization = Execution time (E) / Period (P)

- For multiple or n no. of task
  Total utilization

- A good Scheduling system aims for maximum (near 100%) CPU utilization

Types of Task Scheduling -

- Task Scheduling ~~is~~ is classified into
- Clock - Driven (static scheduling)
- Event - Driven (Dynamic Scheduling)

⟹ Clock Driven Scheduling -

- Based on fixed clock interrupt
- Schedule prepare in advance
- Suitable for periodic task

Types →

a) Table driven scheduling -
- fixed Schedule prepared
- Repeat after a Major cycle
- Not flexible

b) Cyclic Scheduling
- Major cycle divided into Smaller frames
- Task assigned to frames.
- More flexible the travel driven

⇒ Event Driven Scheduling —

- Scheduling decision made when task arrive
- Based on priority
- More flexible

a] Foreground - Background Scheduling →

- Foreground - High priority
- Background - low priority
- Back ground run only when no urgent task.

b] Rate Monotonic Scheduling —

- Fixed priority.
- Shorter period → high priority
- Use for periodic real time System.

c] Earliest Deadline first,

- Dynamic Scheduling
- Task with nearest deadline get higher priority
- Very effecient in real time System

Preemitive vs non preemitive Scheduling.

→ Primitive Scheduling.
- CPU can interrupt low priority task
- Good for real time System.
- Fast response ~~more~~ ~~complex~~.
- More complex

⇒ Non primitive Scheduling →
- Task run until completion.
- Simple & predictable.
- Not Suitable for urgent task.

~~Task vs Process vs THREAD~~

→ task — unit of work to be Scheduled

Primitive →

Example → A Doctor stops filling form to attend an emergency patient

Advantage →
- faster respons to high priority
- Suitable for real time deadline

Disadvantage
- More complex
- Overhead due to contex Switchin

Non primitive →

Ex → Cashier finishes billing one customer before helping the next – even one is in hurry

Adv →
- Simple & predictable
- No interruption & Data conflict

Dis adva →
- urgent task may be delayed.
- poor real time performance.

Task vs Process vs Thread

Task – a basic unit of work (used in embedded System).

Process → A full program in execution with it own memory space.

Thread → A smaller unit inside a process that shares memory.

Bus Structure

Introduction →

- A Bus is communication System used to transfer data b/w different component of a computer System.

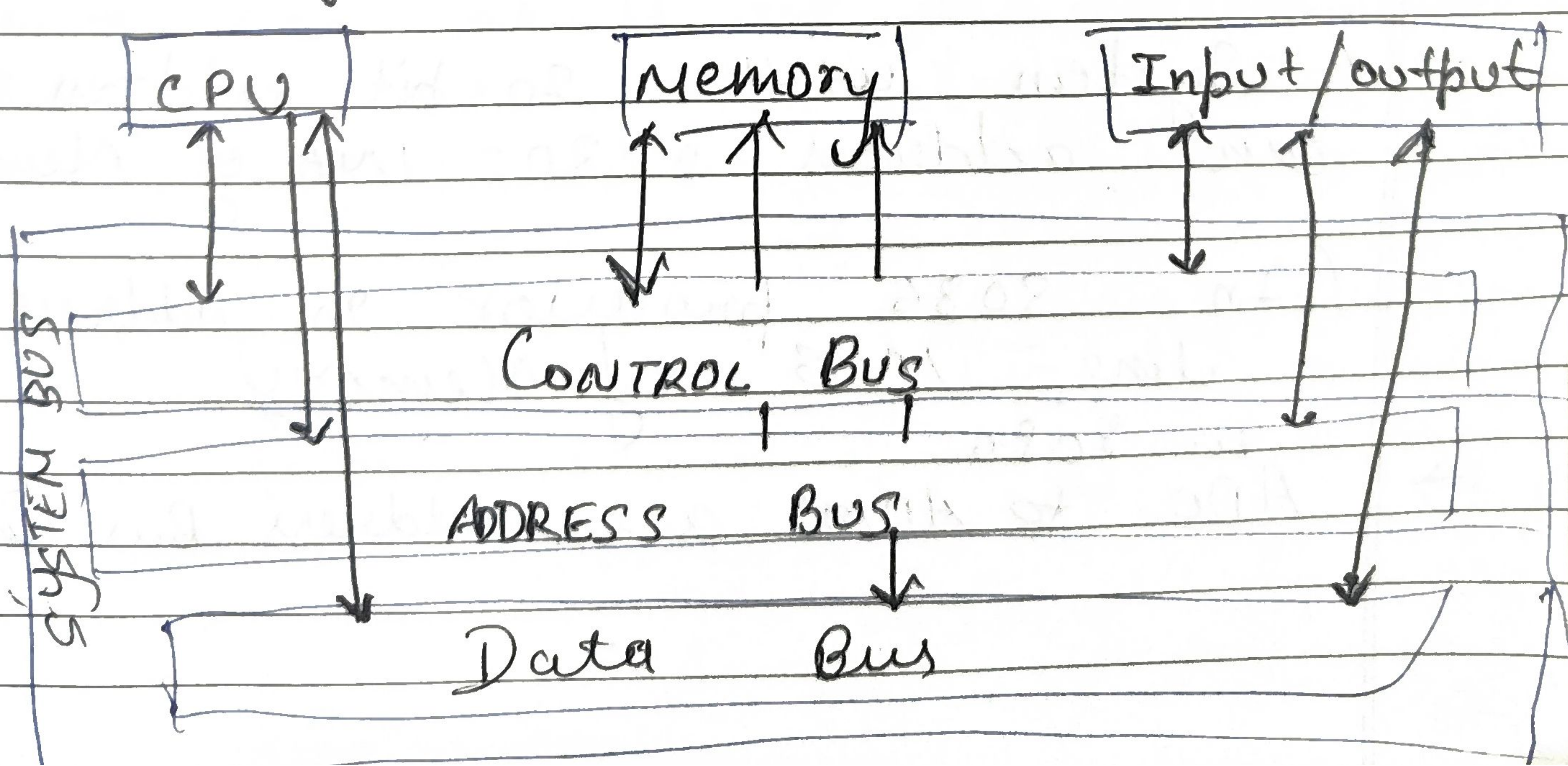
→ It connects: CPU, Main memory, I/o devices

→ Instead of every component having Separate connection with every other component, The bus shared pathway for communi-cation.

→ It simplify data transfer & improve efficiency.

→ It consist of physical connect like wires circuit. or cables

Types of Buses



ADRESS Bus →

→ ~~It is used for identify~~ collection of wire

→ It is a collection of wire used to identify a particular location in main memory

→ It carries the address of main memory locate that the CPU want to access

→ It transport memory address to processor for read & write.

→ The address bus is unidirectional
CPU ⟶ MEMORY / I/O device
E.

→ The Size of address bus determine how many unique location memory can be addressed.

→ A System with 4 bit address $2^4 = 16$ Bytes

→ A System with 20 bit address bus can address 2^{20} = 1MB of Memory

(In 8086 processor 20 address line = 1MB of Memory
"In 8086
→ AD0 to AD19 are address Bus line

DATA BUS →

- Data Bus carries the actual data & Instruction b/w : CPU, Memory, I/O device

→ Data lines carry binary data b/w the CPU, Memory & I/O ~~&~~ bidirectional.

→ Size of Bus determine how much data can transfer at one ~~at~~ ~~a~~ time.

→ ~~16 bit bus can transmit 16 bit of data at a time~~

→ Exa →
16 bit can transmit 16 bits of data at a time.

→ In 8086 processor it is 16 bit processor, the data bus consist of 16 lines from AD0 to AD15 (multiplexed Bus)

→ AD0 to AD15, at T_1 act as address bus & at T_2 act as Data bus.

Control Bus

→ Control Bus carries control & timing Signal b/w CPU, Main Memory & I/O device

→ The Main objective control Bus tells what operation (read or write) perform & when.

→ control Bus also Manage arbitration allowing multiple Bus to communicate efficiently without conflict

→ Control Bus is Bidirectional, Data can flow both dirⁿ from CPU to memory or memory to the CPU

→ E.g. 8086 Microprocessor all other pins are control Bus pin except Vss & Vcc

Bus Architecture →

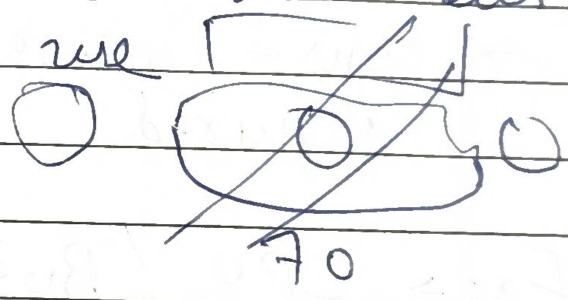
→ Bus architecture defines how Buses are organised in a computer System.

→ It determine how, CPU, Memory & I/O device communicate.

→ It affect, System performance, Scalabilite cost, complexity.

→ Single Bus Architecture →

- All component share one common bus - CPU, Memory, & I/O devices use the Same communication path.



- Only one device can use the bus at a time

- Working principle
- ~~Control~~ CPU places address on bus.
- control Signal are Sent.
- Data is transferred.
- other devices must wait unit bus ~~at a time~~ become free

- Pros - low cost, Simple design, easy to implement
- Cons - Bus Bottleneck, limited performance poor Scalability.

Exa → Intel 8085 & 8086

Heirachichal Bus Structure →

→ It uses multiple buses arranged in levels which Separate high Speed & low speed communication.

→ Working principle →

- CPU communication with memory via high Speed bus.
- I/O device use Separate bus.
- Bridge connect defferent bus
- Multiple transfer can occur efficiently

→ pros- Higher performance, Reduce bottleneck Better Scalability, support multiple devices

→ cons- More complex design, Higher cost Required bus controller

Exa→ CPU Bus, USB Bus, SATA Bus etc.

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Bus Master →

- In Bus System, Device that control the Bus
→ only one master can control one bus at a time

→ It initial read / write operation. Places address on address bus.
→ Generate control Signals & control data transfers

Eg → CPU, DMA, Bus controllers.

Bus SLAVES –

In Bus system, Device that respond to master.
→ it Sends or recieve data. But cannot initiate communication

E.g. RAM, ROM, I/O Device.

Bus Arbitration

- It is process of selecting one master among multiple requester to continue the Bus
→ If multiple master ~~try to~~ use the bus Simultaneously can cause data corruption & System instability

→ It ensure fair access, no conflict & organized communication.

→

→ Centralized - Arbitration →
A Single bus arbiter perform the required arbitration

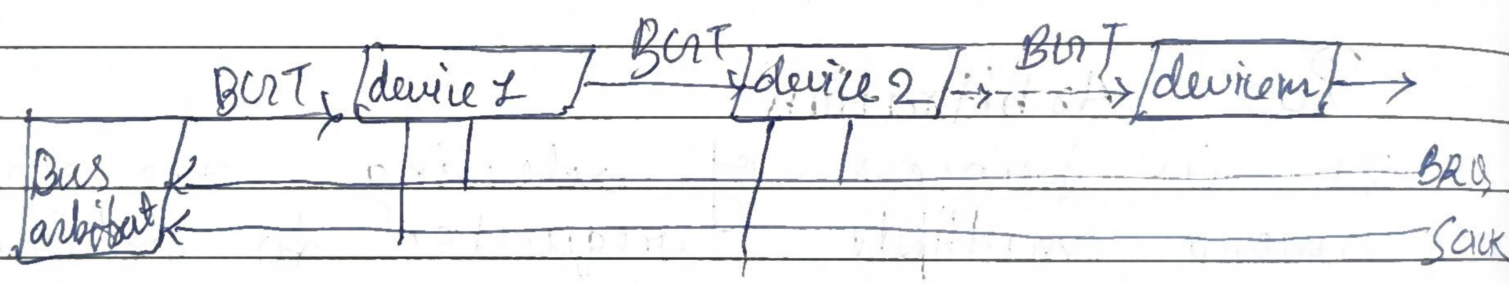
→ Distributed arbitration →
All device participating in the Selection of next bus Master.

→ Method of Centralized bus arbitration

- There are 3 Bus arbitration method

Daisy chain Method

→ It is Method where all the bus master use Same bus.
→ It is method used to decide which device get control of shared System bus



How it work

- BRQ - • Bus request from device.
  • Sent by device to bus arbiter
  • Meaning - I want to use the bus

i BGT - Bus grant Sent by arbiter
- Passed from device to device
Meaning - U are allowed to use the bus

SACK - Acknowledge
- Sent back to arbiter
- Confirm that Some device has taken control.